

Day 2 Assignment: Motherboard & CPU

Front Panel Connectors

Task 1:

ATX Motherboard Labeled Diagram

SATA
SLOT

PCIe SLOT

CMOS
SHELL



Task 2:

Popular CPU Sockets and Compatible Processors

Here are three popular CPU socket types used in modern computer hardware and a compatible CPU model for each:

- 1. Socket AM4 (AMD):**

- * Compatible CPU: AMD Ryzen 5 3600 (Supports Ryzen 1000 to 5000 series desktop processors).**

- 2. Socket LGA 1200 (Intel):**

- * Compatible CPU: Intel Core i5-10400 (Supports Intel's 10th and 11th Generation Core processors).**

- 3. Socket LGA 1700 (Intel):**

- * Compatible CPU: Intel Core i7-12700K (Supports Intel's 12th, 13th, and 14th Generation Core processors).**

Task 3:

Role of CPU and Chipset (Spotify Example)

- **CPU (Central Processing Unit):** It acts as the brain of the computer. It handles all the complex logic, mathematical calculations, and executes the main instructions for any software running on the PC.
- **Motherboard Chipset:** It acts as a communication hub or "traffic controller." It manages how data flows between the CPU, RAM, storage drives, and other external USB or audio devices.

Real-World Example (Opening Spotify):

When you double-click the Spotify app, the CPU processes the command to open the software. However, it is the Chipset that acts as the messenger, fetching the Spotify app files from your storage (SSD/HDD) and sending them to the RAM. When you play a song, the CPU handles the app's interface, while the Chipset routes the actual audio data to the motherboard's sound chip so you can hear the music.

Task 4 :

Identifying Ports and Slots on a Real Motherboard

Below is the photo of a real desktop motherboard where I have identified and labeled 5 key ports and slots:

- **1. CPU Socket:** The main slot located in the center-bottom area where the processor is installed.
- **2. DDR RAM Slots:** The long green vertical slots on the right side used for installing system memory.
- **3. PCIe x16 Slot (Graphics Card Slot):** The main long horizontal slot above the RAM area where a dedicated GPU is connected.
- **4. ATX 24-Pin Power Connector:** The large white rectangular block on the left of the RAM slots that supplies main power from the SMPS.
- **5. SATA Ports:** The four small black blocks located in the top-left corner used to connect storage drives like HDD or SSD.

